

- 7 (a) Fig. 7.1 is a diagram representing a synapse between a chemoreceptor cell from a human taste bud and a dendrite of a sensory neurone.

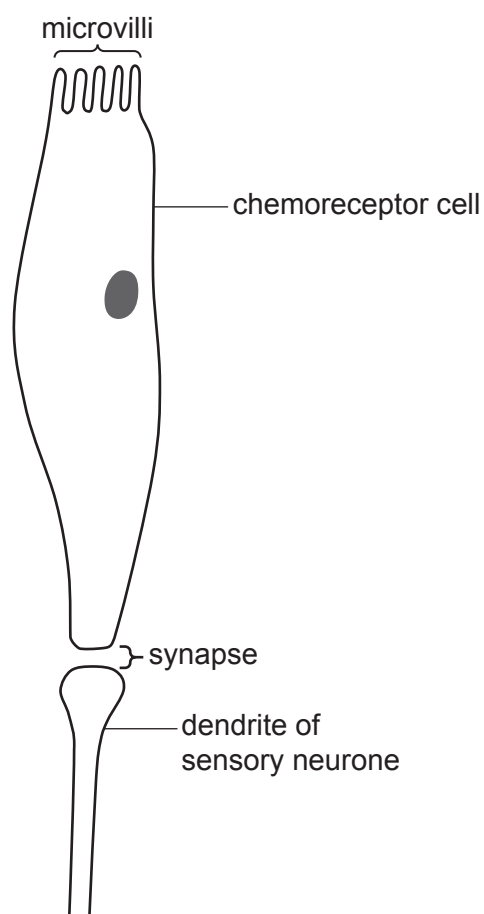


Fig. 7.1

In an experiment, different concentrations of sodium chloride solution were applied to the microvilli of the chemoreceptor cell. The membrane potential of the chemoreceptor cell and the membrane potential of the dendrite of the sensory neurone were recorded for each concentration.

The resting potential of this chemoreceptor cell is -50 mV and the resting potential of the dendrite of this sensory neurone is -70 mV .

The results are shown in Table 7.1.

Table 7.1

concentration of sodium chloride solution/ g dm^{-3}	membrane potential/ mV	
	chemoreceptor cell	dendrite of sensory neurone
0.1	-50	-70
1.0	$+30$	$+40$
10.0	$+30$	$+40$

[4]

[3]

[Total: 7]